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S2**Started on** Monday, 24 August 2026, 5:34 PM**State** Finished**Completed on** Monday, 24 August 2026, 5:35 PM**Time taken** 1 min 2 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int count = 0;
8
9      int rev = 0;
10     count++; // rev = 0 initialization
11
12     int remainder;
13
14     while (n != 0) {
15         count++; // n != 0 condition check (true)
16
17         remainder = n % 10;
18         count++; // remainder = n % 10
19
20         rev = rev * 10 + remainder;
21         count++; // rev = rev * 10 + remainder
22
23         n /= 10;
24         count++; // n /= 10
25     }
26     count++; // final n != 0 condition check (false)
27
28     count++; // print(rev) statement execution
29
30     printf("%d\n", count);
31
32     return 0;
33 }
```

	Input	Expected	Got	
✓	12	11	11	✓

	Input	Expected	Got	
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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